

Ex.no:	DATA VISUALIZATION USING POWER BI

Aim:

To Load the data into PowerBI and transform the data and visualize it to get insights

Algorithm:

Algorithm: Sales Dashboard Calculation

Step 1: Input Data Loading

- Read the dataset from datas.csv containing columns Product, Region, and Sales.
- Dataset rows:
 - Laptop, North, 50000
 - Mobile, South, 35000
 - Printer, East, 25000
 - Laptop, West, 45000
 - Mobile, North, 40000
 - Printer, South, 30000

Step 2: Region Filter Check

- Check slicer selection for Region.
- Default selected regions: East, North, South, West.
- Retain all rows matching the selected regions in Filtered_Data.

Step 3: KPI Total Sales Calculation

- Sum all values in the Sales column from Filtered_Data.
- Total Sales = $50000 + 35000 + 25000 + 45000 + 40000 + 30000 = 225000$ (displayed as 225K).

Step 4: Bar Chart Calculation (Sum of Sales by Product)

- Group Filtered_Data by Product and sum Sales for each product:
 - Laptop Sales = $50000 \text{ (North)} + 45000 \text{ (West)} = 95000$
 - Mobile Sales = $35000 \text{ (South)} + 40000 \text{ (North)} = 75000$
 - Printer Sales = $25000 \text{ (East)} + 30000 \text{ (South)} = 55000$

- Sort products in descending order by total sales: Laptop, Mobile, Printer.

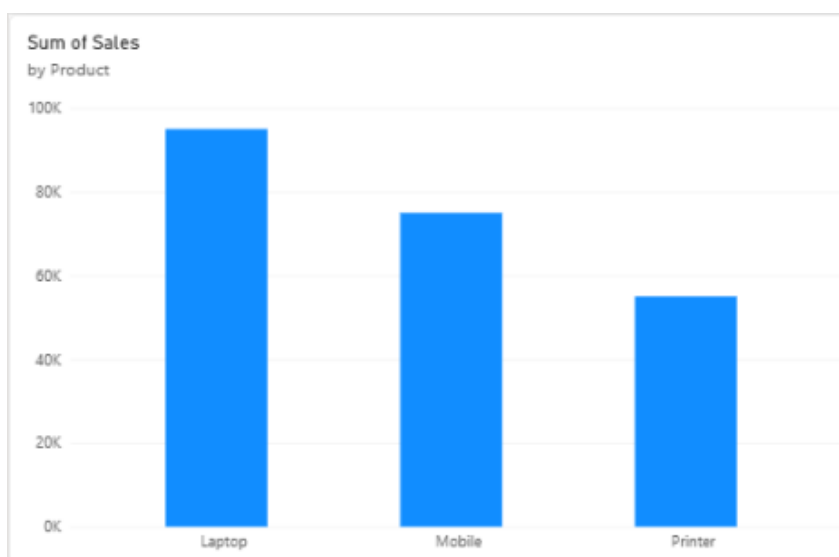
Step 5: Donut Chart Calculation (Sum of Sales by Region)

- Group Filtered_Data by Region and sum Sales for each region:
 - North Sales = $50000 + 40000 = 90000$
 - South Sales = $35000 + 30000 = 65000$
 - West Sales = 45000
 - East Sales = 25000
- Calculate percentage share for each region relative to Total Sales (225000):
 - North Share = $(90000 / 225000) * 100 = 40.00\%$
 - South Share = $(65000 / 225000) * 100 = 28.89\%$
 - West Share = $(45000 / 225000) * 100 = 20.00\%$
 - East Share = $(25000 / 225000) * 100 = 11.11\%$

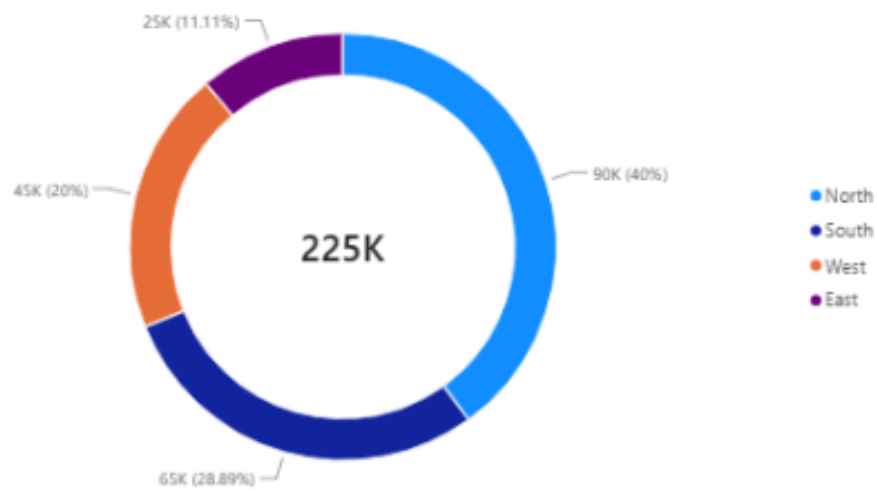
Step 6: Dashboard Render

- Render KPI card with formatted total value 225K.
- Render vertical bar chart with Laptop at 95K, Mobile at 75K, Printer at 55K.
- Render donut chart with region segments North (90K, 40%), South (65K, 28.89%), West (45K, 20%), East (25K, 11.11%), and center value 225K.
- Render region slicer checkboxes for East, North, South, West.
- On slicer selection update, filter rows and recompute steps 2 through 6.

Output:



Sum of Sales
by Region

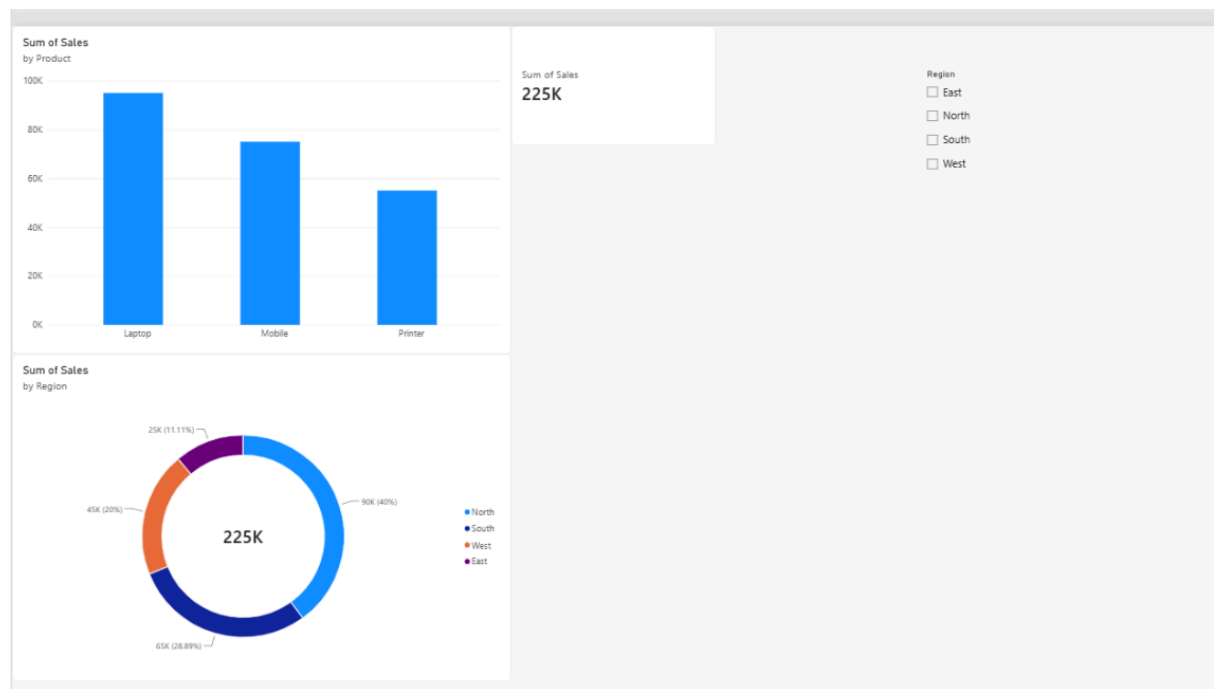


Sum of Sales
225K



Region

- ☐ East
- ☐ North
- ☐ South
- ☐ West



Result: